Decoding Typed Sentences from Non-Invasive Brain Recordings: A Hybrid Conv-Conformer and a NeMo-Style Mamba Architecture for the SpanishBCBL MEG Dataset

COM865 Research Project Proposal — Brain2Qwerty Extension

August 2026 — baseline reproduced on Kelvin2 HPC (test CER 0.389, matching the published V1 baseline of ~0.38)

1. Background and Motivation

Brain2Qwerty (Lévy et al., 2025) demonstrated that sentences typed on a QWERTY keyboard can be decoded from non-invasive magnetoencephalography (MEG) with a character error rate (CER) of roughly 0.32–0.38, using a convolutional encoder followed by a sentence-level transformer trained with per-keystroke cross-entropy. The underlying SpanishBCBL dataset (BCBL, San Sebastián) contains MEG recordings of 35 healthy native Spanish speakers typing briefly memorised 5–8 word sentences (~5.1K sentences, ~193K characters), together with a parallel EEG cohort (~4K sentences). Meta has since released a second-generation codebase (brain2qwerty_v2) targeting the companion English dataset, which replaces the per-keystroke classification framing with a whole-sentence, CTC-based, speech-recognition-style pipeline augmented by word-level contrastive alignment and a LoRA-adapted LLM decoder.

As the foundation for this project, the published V1 baseline has been successfully reproduced on the Kelvin2 HPC cluster (4× V100, DDP): test CER 0.389 (micro-averaged) against the paper's ~0.38, best validation CER 0.381, on 81 usable MEG recordings across 19 held-out test sentences per subject. This proposal builds on that validated baseline in two directions: **Architecture A**, a principled combination of the strongest components of V1 and V2; and **Architecture B**, a novel sequence model in which the attention-based sequence stack is replaced by a NeMo-style hybrid of Mamba-2 state-space blocks with sparse global attention layers.

2. Code-Level Comparison: V1 vs V2

Table 1 summarises the differences extracted directly from the two released packages (`brain2qwerty_v1` and `brain2qwerty_v2`). The two versions are not incremental refinements of one model; they frame the decoding problem differently — per-keystroke classification over fixed 0.5 s windows (V1) versus alignment-free sequence-to-sequence transcription of the whole sentence segment (V2).

Component	V1 (SpanishBCBL)	V2 (EnglishBCBL)
Input unit	0.5 s window per keystroke (start –0.2 s), 50 Hz, bandpass 0.1–20 Hz, baseline correction	Whole-sentence segment, 100 Hz, bandpass 0.5–45 Hz + 50 Hz notch, no baseline

Component	V1 (SpanishBCBL)	V2 (EnglishBCBL)
Channel merger	Per-subject 2D-Fourier merger → 270 virtual channels	Same per-subject 2D-Fourier merger (unchanged)
Encoder	SimpleConvTimeAgg: depth 8, hidden 2048, kernel 3, attention time-pooling → one embedding per keystroke	SimpleConv: depth 4, hidden 1500, kernel 5, per-frame output (no time pooling)
Temporal compression	None (one window = one token)	Learned strided downsampling, kernel 16 / stride 4
Sequence model	TransformerEncoder over keystroke embeddings, 4 layers / 2 heads, ALiBi	Conformer, 4 layers / 4 heads, FFN 1024, depthwise conv 17, group norm
Auxiliary pathway	None	Auxiliary CTC head whose softmax is linearly blended back into the Conformer input
Training objective	Per-keystroke cross-entropy (29 classes), labels from logged key timestamps	Staged 3-loss schedule: char-level CTC (aux+final blend, α=0.7) from epoch 0; word-level SigLIP contrastive (DTW-matched to frozen TinyLlama token embeddings) from epoch 150; LLM loss from epoch 225
Augmentation	None	On-device: per-channel constant offset (SD 0.3), SpecAugment time/frequency masking, time-stretch
Decoder	Argmax per keystroke; optional offline KenLM n-gram beam rescoring	TinyLlama-1.1B + LoRA (rank 2), conditioned on CTC greedy text and neural word embeddings; beam 16, length penalty 0.2
Optimiser	AdamW lr 5e-5, wd 1e-4, OneCycleLR	AdamW lr 8e-4, wd 1e-3, warmup + cosine, bf16, grad-accum 2
Metrics	CER, WER (argmax keystrokes)	CTC-CER (greedy, blank-collapsed), LLM CER/WER, SemER (RoBERTa-large semantic error rate, test only)

Table 1 — V1 vs V2, extracted from the released configs and module code.

Three observations motivate the proposed architectures. **(i)** V1's encoder is the stronger front-end — deeper (8 vs 4 layers), wider (2048 vs 1500), with attention-based temporal pooling — but its per-keystroke framing hard-codes the logged keypress timestamp as ground truth, discards all neural activity outside the 0.5 s windows, and cannot recover when the label timing is noisy. **(ii)** V2's CTC objective removes the timing assumption and its staged contrastive + LLM losses inject linguistic structure, but its encoder is shallower and V2 has only been applied to the English cohort. **(iii)** Both versions rely on attention as the sole long-range sequence mixer, whose quadratic cost and fixed positional biases are a poor match for the irregular inter-keystroke intervals of natural typing.

3. Architecture A — B2Q-Hybrid (Best of Both Worlds)

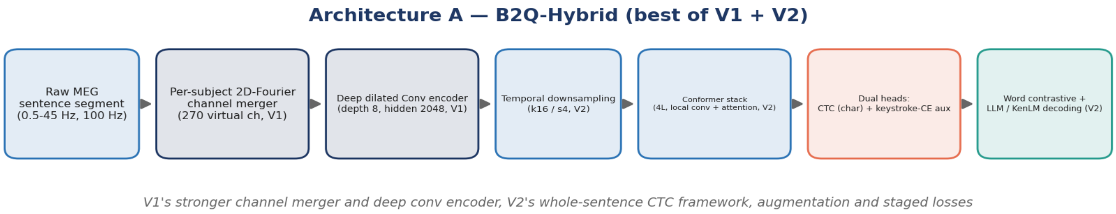

Figure 1 — Architecture A data flow.

Architecture A ports the V2 training framework to SpanishBCBL while restoring V1's stronger front-end. Concretely: whole-sentence MEG segments are preprocessed with V2's settings (100 Hz, 0.5–45 Hz, notch 50 Hz) and V2's on-device augmentations; the per-subject 2D-Fourier merger (retained by both versions) feeds V1's deep dilated convolutional encoder (depth 8, hidden 2048) with per-frame outputs instead of V1's attention pooling; V2's learned temporal downsampling and Conformer stack model the sentence sequence. Training uses a dual objective: the CTC loss (with V2's auxiliary head and blending) as the primary signal, plus V1's per-keystroke cross-entropy re-introduced as an auxiliary alignment loss at CTC-segmented keystroke positions, giving the encoder the explicit motor-event supervision that made V1 effective. The word-level SigLIP contrastive loss and LLM decoding stage follow V2's staged schedule; the cheaper KenLM n-gram rescoring from V1 is retained as a lightweight decoder baseline.

Hypothesis A: combining V1's encoder capacity and keystroke supervision with V2's alignment-free CTC framework and linguistic losses yields a lower CER on SpanishBCBL than either released model family, because the two training signals are complementary (explicit motor alignment vs. free alignment plus language structure).

4. Architecture B — B2Q-Mamba (NeMo-Style Hybrid)

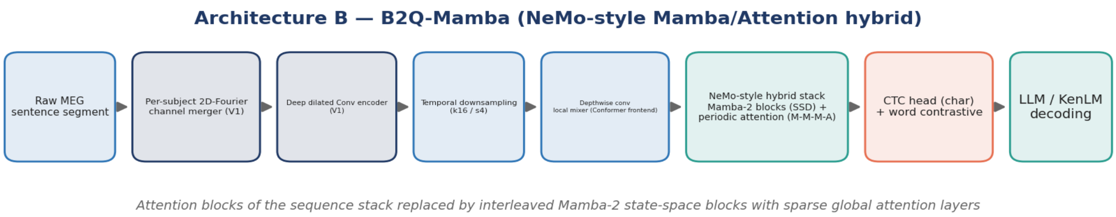

Figure 2 — Architecture B data flow.

Architecture B keeps Architecture A's front-end and losses but replaces the attention-based sequence stack (V2's Conformer / V1's sentence transformer) with a NeMo-style hybrid Mamba–Transformer block pattern. Mamba-2 (structured state-space duality, SSD) blocks perform linear-time selective sequence modelling: their input-dependent state transitions are a natural fit for typing signals, where the relevant neural events occur at irregular, subject-specific intervals rather than on a fixed clock. Following the Nemotron-H / NeMo hybrid design, Mamba-2 blocks are interleaved with periodic global self-attention layers in an M-M-M-A pattern (three Mamba blocks per attention layer), so that cheap selective state-space mixing handles the bulk of the temporal structure while a small number of attention layers provide exact long-range retrieval across the sentence. The Conformer's depthwise convolution is kept as the local mixer in front of the hybrid stack, preserving the local feature smoothing that both released versions rely on.

Expected advantages over the pure-attention stacks: (i) linear scaling in sentence length, allowing longer context (multiple sentences, or cross-sentence language context) at the same compute budget; (ii) an inductive bias toward smooth, event-driven dynamics matching the motor-production signal; (iii) fewer attention layers means fewer positional-bias assumptions on an irregularly sampled event sequence. Risks: Mamba blocks are newer and less validated on small neural datasets (~81 recordings); state-space layers can underfit rare long-range dependencies, which the periodic attention layers are designed to mitigate.

Hypothesis B: a hybrid Mamba-2/attention sequence stack matches or improves CER relative to the Conformer stack at equal parameter budget on SpanishBCBL, with measurably better compute scaling on longer sequences.

5. Experimental Plan and Evaluation Protocol

5.1 Staged experiments

Stage	Experiment	Comparison / decision
S0	Run V1 KenLM n-gram rescoring on the reproduced baseline (pipeline exists, not yet executed)	Establishes the LM-corrected reference CER; zero training cost
S1	Architecture A ablations on SpanishBCBL: (a) V2 framework as-is ported to Spanish; (b) + V1 encoder; (c) + keystroke-CE auxiliary loss	Isolates the contribution of each merged component against the 0.389 CER baseline
S2	Architecture B: swap the Conformer for the M-M-M-A Mamba-2 hybrid at matched parameter count and training schedule	Hypothesis B test: CER parity or better, plus wall-clock scaling on longer contexts
S3	Decoder study on the best encoder: KenLM vs LoRA-LLM (Spanish LM, e.g. a Spanish Llama-family model) vs none	Quantifies the neural vs. linguistic share of the error reduction
S4	Per-subject analysis and adaptation (fine-tune on worst subjects, e.g. S20 at CER 0.552)	Addresses the dominant error source observed in the reproduction (subject-level spread 0.287–0.552)

Table 2 — Staged experimental plan.

5.2 Metrics and protocol

Primary metric: CER (micro-averaged, matching the paper and the reproduced baseline), reported both with and without language-model decoding. Secondary metrics: WER, exact-sentence rate, and V2's SemER (RoBERTa-large semantic error rate) for the LLM decoding stage; per-subject CER distributions throughout, since the reproduction showed subject identity is the dominant variance source. All models use the V1 deterministic subject-disjoint split on SpanishBCBL for direct comparability with the reproduced baseline, with the same early-stopping criterion (val CER, patience 30). Each configuration is trained on 4× V100 with the V2 bf16/grad-accum settings; the 4.5 GB feature cache pipeline is reused, with cache rebuilt at 64 GB RAM to avoid the OOM corruption observed during the reproduction.

5.3 Expected outcomes and risks

Expected outcome: Architecture A improves over 0.389 CER (target: match or beat the paper's LM-corrected ~0.32) because each merged component addresses a known, independently verified weakness; Architecture B provides the first evaluation of state-space sequence models for MEG typing decoding, publishable whether or not it wins, provided the comparison is at matched parameter budget. Main risks and mitigations: small dataset (mitigated by V2 augmentations, per-subject merger, and the staged-loss schedule which delays the data-hungry LLM stage); Mamba implementation risk (mitigated by using the reference mamba-ssm kernels and keeping Architecture A as a fallback contribution); Spanish LLM availability for the V2-style decoding stage (mitigated by KenLM Spanish n-grams as the primary linguistic decoder, with the LLM stage optional).

6. Indicative Timeline

Weeks	Work package
1–2	S0 rescoring; port V2 data pipeline and CTC framework to SpanishBCBL; rebuild feature cache
3–5	Architecture A training and ablations (S1)
6–8	Architecture B implementation (Mamba-2 hybrid stack) and matched-budget training (S2)
9–10	Decoder study and per-subject adaptation (S3, S4)
11–12	Analysis, figures, and final report writing

Table 3 — Twelve-week plan on Kelvin2 HPC.

References

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